
30 Polynomials and the FFT

The straightforward method of adding two polynomials of degree n takes $\Theta(n)$ time, but the straightforward method of multiplying them takes $\Theta(n^2)$ time. This chapter will show how the fast Fourier transform, or FFT, can reduce the time to multiply polynomials to $\Theta(n \lg n)$.

The most common use for Fourier transforms, and hence the FFT, is in signal processing. A signal is given in the *time domain*: as a function mapping time to amplitude. Fourier analysis expresses the signal as a weighted sum of phase-shifted sinusoids of varying frequencies. The weights and phases associated with the frequencies characterize the signal in the *frequency domain*. Among the many everyday applications of FFT's are compression techniques used to encode digital video and audio information, including MP3 files. Many fine books delve into the rich area of signal processing, and the chapter notes reference a few of them.

Polynomials

A *polynomial* in the variable x over an algebraic field F represents a function $A(x)$ as a formal sum:

$$A(x) = \sum_{j=0}^{n-1} a_j x^j .$$

The values $a_0, a_1, \dots, a_{n-1}$ are the *coefficients* of the polynomial. The coefficients and x are drawn from a field F , typically the set $\mathbb{C}$ of complex numbers. A polynomial $A(x)$ has *degree* k if its highest nonzero

coefficient is a_k , in which case we say that $\text{degree}(A) = k$. Any integer strictly greater than the degree of a polynomial is a **degree-bound** of that polynomial. Therefore, the degree of a polynomial of degree-bound n may be any integer between 0 and $n - 1$, inclusive.

A variety of operations extend to polynomials. For **polynomial addition**, if $A(x)$ and $B(x)$ are polynomials of degree-bound n , their **sum** is a polynomial $C(x)$, also of degree-bound n , such that $C(x) = A(x) + B(x)$ for all x in the underlying field. That is, if

$$A(x) = \sum_{j=0}^{n-1} a_j x^j \text{ and } B(x) = \sum_{j=0}^{n-1} b_j x^j ,$$

then

$$C(x) = \sum_{j=0}^{n-1} c_j x^j ,$$

where $c_j = a_j + b_j$ for $j = 0, 1, \dots, n - 1$. For example, given the polynomials $A(x) = 6x^3 + 7x^2 - 10x + 9$ and $B(x) = -2x^3 + 4x - 5$, their sum is $C(x) = 4x^3 + 7x^2 - 6x + 4$.

For **polynomial multiplication**, if $A(x)$ and $B(x)$ are polynomials of degree-bound n , their **product** $C(x)$ is a polynomial of degree-bound $2n - 1$ such that $C(x) = A(x)B(x)$ for all x in the underlying field. You probably have multiplied polynomials before, by multiplying each term in $A(x)$ by each term in $B(x)$ and then combining terms with equal powers. For example, you can multiply $A(x) = 6x^3 + 7x^2 - 10x + 9$ and $B(x) = -2x^3 + 4x - 5$ as follows:

$$\begin{array}{r}
 6x^3 + 7x^2 - 10x + 9 \\
 - 2x^3 \qquad \qquad + 4x - 5 \\
 \hline
 - 30x^3 - 35x^2 + 50x - 45 \quad (\text{multiply } A(x) \text{ by } -5) \\
 24x^4 + 28x^3 - 40x^2 + 36x \quad (\text{multiply } A(x) \text{ by } 4x) \\
 - 12x^6 - 14x^5 + 20x^4 - 18x^3 \quad (\text{multiply } A(x) \text{ by } -2x^3) \\
 \hline
 - 12x^6 - 14x^5 + 44x^4 - 20x^3 - 75x^2 + 86x - 45
 \end{array}$$

Another way to express the product $C(x)$ is

$$C(x) = \sum_{j=0}^{2n-2} c_j x^j , \tag{30.1}$$